

Electron Diffraction

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1. RESULT AND DISCUSSION

Figure 1 shows that the slope of the inner ring data fit is $m_1 = 3.14241 \pm 0.198105$ and of the outer ring is $m_2 = 6.21786 \pm 0.199757$. The ratio between m_1 and m_2 is 1.97, indicating that the diffraction pattern is caused by the variation of carbon graphite diffraction grid spacing instead of the diffraction order $n=1, 2, 3...$ This is because if the order difference of diffraction n causes the diffraction pattern, the ratio should be an integer of two or higher(if applicable). The d_1 corresponds to the m_2 , and d_2 corresponds to the m_1 , differences diffraction order $n=1$. The values of h are $h_1 = 7.22313 \pm 0.45593 \times 10^{-34}$ and $h_2 = 8.25414 \pm 0.26548 \times 10^{-34}$

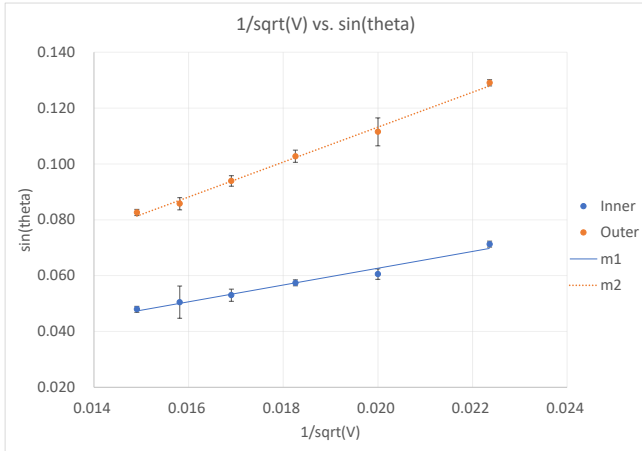


Figure 1. Using the Least Squares Fit, the best-fit m_1 slope is 3.14241 with uncertainty 0.198105. The best-fit m_2 slope is 6.21786 with uncertainty 0.199757

2. DATA AND ANALYSIS

2.1. Data

In the experiment, six voltage values with the corresponding diameter were collected. Range from 2.00kV to 4.50kV, with 0.50kV interval. Three trials were conducted and represented the average $\pm$ standard deviation.

Voltage (kV)	r_inner(cm)	r_outer(cm)
4.50	1.27 ± 0.03	2.18 ± 0.03
4.00	1.33 ± 0.06	2.27 ± 0.15
3.50	1.40 ± 0.05	2.48 ± 0.06
3.00	1.52 ± 0.06	2.72 ± 0.03
2.50	1.60 ± 0.13	2.95 ± 0.05
2.00	1.88 ± 0.03	3.42 ± 0.03

2.2. Analysis

In the wave of electrons, we can set the wavelength λ by setting the accelerator voltage V . V_e is the kinetic energy converted from the potential of the electron. So, we can adjust the wavelength by adjusting the voltage. By the Bragg scattering formula,

$$2d\sin\theta = n\lambda$$

We set up the measurement. For each potential setting, y is $\sin\theta$, x is $V^{-\frac{1}{2}}$, and m is $\frac{1}{d} \frac{nh}{\sqrt{8m_e e}}$.

The n is the order of diffraction. The d is the graphite atom spacing(which is the diffraction grid spacing) The θ is the diffraction angle.

$$\lambda = \frac{h}{p}$$

$$eV = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m} = \frac{h^2}{\lambda^2 2m}$$

$$\lambda = \sqrt{\frac{h^2}{\lambda^2 2meV}}$$

$$\sin\theta = \frac{1}{2d} \frac{nh}{\sqrt{2m_e e}} V^{-\frac{1}{2}} = \frac{1}{d} \frac{nh}{\sqrt{8m_e e}} V^{-\frac{1}{2}}$$

Given the slope of the least square fit line, Planck's Constant can be obtained by the following equation:

$$m_1 = \frac{1}{d_2} \frac{h}{\sqrt{8m_e e}}$$

$$h_1 = m_1 d_2 \sqrt{8m_e e} = 7.22313 \times 10^{-34}$$

Similarly, the second constant can be obtained by:

$$h_2 = m_2 d_1 \sqrt{8m_e} = 8.25414 \times 10^{-34}$$

Taking the uncertainty into consideration, the result of h_1 and h_2 is $h_1 = 7.22313 \pm 0.45593 \times 10^{-34}$ and $h_2 = 8.25414 \pm 0.26548 \times 10^{-34}$ respectively.

3. UNCERTAINTY

The uncertainty of Planck's Constant is calculated by the general formula of uncertainty. There are 3 terms in the uncertainty propagation: slope Δm , electron mass Δm_e , and electron charge Δe

$$\sigma_f^2 = \sigma^2 x \left(\frac{\partial f}{\partial x} \right)^2 + \sigma^2 y \left(\frac{\partial f}{\partial y} \right)^2 + \sigma^2 z \left(\frac{\partial f}{\partial z} \right)^2 \dots$$

The following equation represents the calculation of h:

$$h(m, e) = md\sqrt{8m_e e}$$

The partial derivative of slope m uncertainty term and electron mass is:

$$\Delta m = d\sqrt{8m_e e}$$

$$\Delta m_e = 0.0000000028 \times 10^{-31} kg$$

We didn't take the electron charge uncertainty term Δe into calculation because the NIST provides the uncertainty = exact, which is set to zero here. The Δm_e term is 1.1×10^{-146} , which is neglectable.

$$\sigma_e^2 \left(\frac{1}{2} md\sqrt{8m_e} e^{-1/2} \right)^2 = 0$$

$$\sigma_{m_e}^2 \left(\frac{1}{2} md\sqrt{8e} m_e^{-1/2} \right)^2 = 1.1 \times 10^{-146} \approx 0$$

Combined with the general formula of uncertainty, the uncertainty of h is finally

$$\sigma_h = \sqrt{\sigma_m^2 (d\sqrt{8m_e e})^2}$$

Therefore, the uncertainty of Planck's Constant is $\sigma_{h_1} = 0.455927 \times 10^{-34}$ is obtained by:

$$\sigma_{h_1} = \sqrt{\sigma_{m_1}^2 (d_2 \sqrt{8m_e e})^2}$$

The $\sigma_{h_2} = 0.265478 \times 10^{-34}$ is obtained by:

$$\sigma_{h_2} = \sqrt{\sigma_{m_2}^2 (d_1 \sqrt{8m_e e})^2}$$

The table of e information above is referred from the NIST website(at nist.gov, see "Physical Reference Data,

Electron Numerical value	$1.602176634 \times 10^{-19} C$
Standard uncertainty	(exact)
Relative standard uncertainty	(exact)

Electron mass Numerical value	$9.1093837015 \times 10^{-31} kg$
Standard uncertainty	$0.0000000028 \times 10^{-31} kg$
Relative standard uncertainty	3.0×10^{-10}

Physical Constants"). The electron mass uncertainty is $0.0000000028 \times 10^{-31} kg$.

Finally, Planck's Constant with uncertainty are:

$$h_1 = 7.22313 \pm 0.45593 \times 10^{-34}$$

$$h_2 = 8.25414 \pm 0.26548 \times 10^{-34}$$